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Beam management in multi-TRP operation

Abstract

A cell performs beam management operations for a user equipment (UE). The cell transmits a first group of multiple reference signals via a first transmission reception point (TRP) to a user equipment (UE), transmits a second group of multiple reference signals via a second different TRP to the UE, receives an indication that the UE has selected a first beam associated with one of the multiple reference signals of the first group and a second beam associated with one of the multiple reference signals of the second group and transmits downlink data to the UE using the first beam and the second beam.

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Background/Summary

BACKGROUND

(1) A user equipment (UE) may establish a connection to at least one of multiple different networks or types of networks. Signaling between the UE and the network may be achieved via beamforming. Beamforming is an antenna technique used to transmit a directional signal which may be referred to as a beam.

(2) A cell of the network may be configured with multiple transmission reception points (TRPs) each configured to perform beamforming. For example, the cell may transmit a first beam from a first TRP to the UE and a second beam from a second TRP to the UE. To acquire and maintain a beam between the UE and each of the TRPs, beam management techniques may be implemented on both the UE side and the network side.

SUMMARY

(3) Some exemplary embodiments are related to a baseband processor configured to perform operations. The operations include transmitting a first group of multiple reference signals via a first transmission reception point (TRP) to a user equipment (UE), transmitting a second group of multiple reference signals via a second different TRP to the UE, receiving an indication that the UE has selected a first beam associated with one of the multiple reference signals of the first group and a second beam associated with one of the multiple reference signals of the second group and transmitting downlink data to the UE using the first beam and the second beam.

(4) Other exemplary embodiments are related to a cell having a communication interface configured to communicate with a user equipment (UE) and a processor communicatively coupled to the communication interface and configured to perform operations. The operations include transmitting a first group of multiple reference signals via a first transmission reception point (TRP) to a user equipment (UE), transmitting a second group of multiple reference signals via a second different TRP to the UE, receiving an indication that the UE has selected a first beam associated with one of the multiple reference signals of the first group and a second beam associated with one of the multiple reference signals of the second group and transmitting downlink data to the UE using the first beam and the second beam.

(5) Still further exemplary embodiments are related to a method performed by a cell of a network. The method includes transmitting a first group of multiple reference signals via a first transmission reception point (TRP) to a user equipment (UE), transmitting a second group of multiple reference signals via a second different TRP to the UE, receiving an indication that the UE has selected a first beam associated with one of the multiple reference signals of the first group and a second beam associated with one of the multiple reference signals of the second group and transmitting downlink data to the UE using the first beam and the second beam.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an exemplary network arrangement according to various exemplary embodiments.
- (2) FIG. 2 shows an example of multiple transmission reception points (TRPs) deployed at different locations.
- (3) FIG. 3 shows an exemplary user equipment (UE) according to various exemplary embodiments.
- (4) FIG. 4 shows a signaling diagram for an exemplary multi-TRP beam reporting procedure according to various exemplary embodiments.
- (5) FIG. 5 illustrates an example of a channel state information reference signal (CSI-RS) group pair.
- (6) FIG. 6 shows a signaling diagram for an exemplary beam failure procedure for multi-TRP operation according to various exemplary embodiments.
- (7) FIG. 7 shows an exemplary cell according to various exemplary embodiments.

DETAILED DESCRIPTION

- (8) The exemplary embodiments may be further understood with reference to the following description and the related appended drawings, wherein like elements are provided with the same reference numerals. The exemplary embodiments relate to beam management for multi-transmission reception point (TRP) operation.
- (9) The exemplary embodiments are described with regard to a UE. However, reference to a UE is merely provided for illustrative purposes. The exemplary embodiments may be utilized with any electronic component that may establish a connection to a network and is configured with the hardware, software, and/or firmware to exchange information and data with the network. Therefore, the UE as described herein is used to represent any appropriate electronic component.
- (10) The exemplary embodiments are also described with regard to a 5G New Radio (NR) network. However, reference to a 5G NR network is merely provided for illustrative purposes. The exemplary embodiments may be utilized with any network that utilizes beamforming. Therefore, the 5G NR network as described herein may represent any type of network that implements beamforming.
- (11) A person of ordinary skill in the art would understand that beamforming is an antenna technique that is utilized to transmit or receive a directional signal. From the perspective of a transmitting device, beamforming may refer to propagating a directional signal. Throughout this description, a beamformed signal may be referred to as a “beam” or a “transmitter beam.” The transmitter beam may be generated by having a plurality of antenna elements radiate the same signal. Increasing the number of antenna elements radiating the signal decreases the width of the radiation pattern and increases the gain. Thus, a transmitter beam may vary in width and be propagated in any of a plurality of different directions.
- (12) From the perspective of a receiving device, beamforming may refer to tuning a receiver to listen to a direction of interest. Throughout this description, the spatial area encompassed by the receiver listening in the direction of interest may be referred to as a “beam” or a “receiver beam.” The receiver beam may be generated by configuring the parameters of a spatial filter on a receiver antenna array to listen in a direction of interest and filter out any noise from outside the direction of interest. Like a transmitter beam, a receiver beam may also vary in width and be directed in any of a plurality of different areas of interest.
- (13) In addition, the exemplary embodiments are described with regard to a next generation node B (gNB) that is configured with multiple TRPs. Throughout this description, a TRP generally refers to a set of components configured to transmit and/or receive a beam. In some embodiments, multiple TRPs may be deployed locally at the gNB. For example, the gNB may include multiple antenna arrays/panels that are each configured to generate a different beam. In other embodiments, multiple TRPs may be deployed at various different locations and connected to the gNB via a

backhaul connection. For example, multiple small cells may be deployed at different locations and connected to the gNB. However, these examples are merely provided for illustrative purposes. Those skilled in the art will understand that TRPs are configured to be adaptable to a wide variety of different conditions and deployment scenarios. Thus, any reference to a TRP being a particular network component or multiple TRPs being deployed in a particular arrangement is merely provided for illustrative purposes. The TRPs described herein may represent any type of network component configured to transmit and/or receive a beam.

(14) The exemplary embodiments relate to implementing beam management techniques on both the UE side and the network side. Beam management generally refers to a set of procedure configured to acquire and maintain a beam between a TRP and the UE. In a first aspect, the exemplary embodiments relate to beam reporting. As will be described in more detail below, on the network side, this may include multiple TRPs each transmitting a set of reference signals to the UE. The UE may collect measurement data using the reference signals, select a beam associated with each TRP and then report the selected beams to the network. In response to the beam reporting, the network may configure the UE with a beam from each TRP. In a second aspect, the exemplary embodiments relate to beam failure detection (BFD) and beam failure recovery (BFR) procedures. BFD generally relates to determining that a serving beam is not providing adequate quality and/or performance in the downlink. BFR generally relates to assisting the network with scheduling subsequent downlink communications using a different beam that is likely to provide adequate quality and/or performance in the downlink. The exemplary beam management techniques described herein may be used in conjunction with currently implemented beam management mechanisms, future implementations of beam management mechanisms or independently from other beam management mechanisms.

(15) FIG. 1 shows an exemplary network arrangement **100** according to various exemplary embodiments. The exemplary network arrangement **100** includes a UE **110**. Those skilled in the art will understand that the UE **110** may be any type of electronic component that is configured to communicate via a network, e.g., mobile phones, tablet computers, desktop computers, smartphones, phablets, embedded devices, wearables, Internet of Things (IoT) devices, etc. It should also be understood that an actual network arrangement may include any number of UEs being used by any number of users. Thus, the example of a single UE **110** is merely provided for illustrative purposes.

(16) The UE **110** may be configured to communicate with one or more networks. In the example of the network configuration **100**, the network with which the UE **110** may wirelessly communicate is a 5G NR radio access network (RAN) **120**. However, the UE **110** may also communicate with other types of networks (e.g. 5G cloud RAN, a next generation RAN (NG-RAN), a long term evolution RAN, a legacy cellular network, a WLAN, etc.) and the UE **110** may also communicate with networks over a wired connection. With regard to the exemplary embodiments, the UE **110** may establish a connection with the 5G NR RAN **120**. Therefore, the UE **110** may have a 5G NR chipset to communicate with the NR RAN **120**.

(17) The 5G NR RAN **120** may be a portion of a cellular network that may be deployed by a network carrier (e.g., Verizon, AT&T, T-Mobile, etc.). The 5G NR RAN **120** may include, for example, cells or base stations (Node Bs, eNodeBs, HeNBs, eNBS, gNBs, gNodeBs, macrocells, microcells, small cells, femtocells, etc.) that are configured to send and receive traffic from UEs that are equipped with the appropriate cellular chip set.

(18) In network arrangement **100**, the 5G NR RAN **120** includes a cell **120A** that represents a gNB that is configured with multiple TRPs. Each TRP may represent one or more components configured to transmit and/or receive a beam. In some embodiments, multiple TRPs may be deployed locally at the cell **120A**. In other embodiments, multiple TRPs may be distributed at different locations and connected to the gNB.

(19) FIG. 2 shows an example of multiple TRPs deployed at different locations. In this example,

the gNB **205** is configured with a first TRP **210** via a backhaul connection **212** and a second TRP **220** via backhaul connection **222**. Each of the TRPs **210**, **220** may transmit a beam to and/or receive a beam from the UE **110**. However, the gNB **205** may be configured to control the TRPs **210**, **220** and perform operations such as, but not limited to, assigning resources, configuring group pairs, configuring reporting restrictions, implementing beam management techniques, etc.

(20) The example shown in FIG. **2** is not intended to limit the exemplary embodiments in any way. Those skilled in the art will understand that 5G NR TRPs are adaptable to a wide variety of different conditions and deployment scenarios. An actual network arrangement may include any number of different types of cells and/or TRPs being deployed by any number of RANs in any appropriate arrangement. Thus, the example of a single cell **120A** in FIG. **1** and a single gNB **205** with two TRPs **210**, **220** in FIG. **2** is merely provided for illustrative purposes.

(21) Returning to the network arrangement **100** of FIG. **1**, the cell **120A** may include one or more communication interfaces to exchange data and/or information with UEs, the corresponding RAN, the cellular core network **130**, the internet **140**, etc. Further, the cell **120A** may include a processor configured to perform various operations. An example of cell **120A** is shown in FIG. **7**. The processor **705** of the cell **120A** may be configured to perform operations related to access barring. However, reference to a processor is merely for illustrative purposes. The operations of the cell **120A** may also be represented as a separate incorporated component of the cell **120A** or may be a modular component coupled to the cell **120A**, e.g., an integrated circuit with or without firmware. For example, the integrated circuit may include input circuitry to receive signals and processing circuitry to process the signals and other information. In addition, in some examples, the functionality of the processor **705** is split among two or more processors such as a baseband processor and an applications processor. The exemplary embodiments may be implemented in any of these or other configurations of a cell.

(22) The UE **110** may connect to the 5G NR-RAN **120** via the cell **120A**. Those skilled in the art will understand that any association procedure may be performed for the UE **110** to connect to the 5G NR-RAN **120**. For example, as discussed above, the 5G NR-RAN **120** may be associated with a particular cellular provider where the UE **110** and/or the user thereof has a contract and credential information (e.g., stored on a SIM card). Upon detecting the presence of the 5G NR-RAN **120**, the UE **110** may transmit the corresponding credential information to associate with the 5G NR-RAN **120**. More specifically, the UE **110** may associate with a specific cell (e.g., the cell **120A**). However, as mentioned above, reference to the 5G NR-RAN **120** is merely for illustrative purposes and any appropriate type of RAN may be used.

(23) In addition to the 5G NR RAN **120**, the network arrangement **100** also includes a cellular core network **130**, the Internet **140**, an IP Multimedia Subsystem (IMS) **150**, and a network services backbone **160**. The cellular core network **130** may be considered to be the interconnected set of components that manages the operation and traffic of the cellular network. The cellular core network **130** also manages the traffic that flows between the cellular network and the Internet **140**. The IMS **150** may be generally described as an architecture for delivering multimedia services to the UE **110** using the IP protocol. The IMS **150** may communicate with the cellular core network **130** and the Internet **140** to provide the multimedia services to the UE **110**. The network services backbone **160** is in communication either directly or indirectly with the Internet **140** and the cellular core network **130**. The network services backbone **160** may be generally described as a set of components (e.g., servers, network storage arrangements, etc.) that implement a suite of services that may be used to extend the functionalities of the UE **110** in communication with the various networks.

(24) FIG. **3** shows an exemplary UE **110** according to various exemplary embodiments. The UE **110** will be described with regard to the network arrangement **100** of FIG. **1**. The UE **110** may include a processor **305**, a memory arrangement **310**, a display device **315**, an input/output (I/O) device **320**, a transceiver **325** and other components **330**. The other components **330** may include,

for example, an audio input device, an audio output device, a power supply, a data acquisition device, ports to electrically connect the UE **110** to other electronic devices, etc.

(25) The processor **305** may be configured to execute a plurality of engines of the UE **110**. For example, the engines may include a multi-TRP beam management engine **335**. The multi-TRP beam management engine **335** may be configured to perform operations related to beam management such as, collecting measurement data, beam selection, beam failure detection, beam failure recovery, etc.

(26) The above referenced engine being an application (e.g., a program) executed by the processor **305** is only exemplary. The functionality associated with the engine may also be represented as a separate incorporated component of the UE **110** or may be a modular component coupled to the UE **110**, e.g., an integrated circuit with or without firmware. For example, the integrated circuit may include input circuitry to receive signals and processing circuitry to process the signals and other information. The engines may also be embodied as one application or separate applications. In addition, in some UEs, the functionality described for the processor **305** is split among two or more processors such as a baseband processor and an applications processor. The exemplary embodiments may be implemented in any of these or other configurations of a UE.

(27) The memory arrangement **310** may be a hardware component configured to store data related to operations performed by the UE **110**. The display device **315** may be a hardware component configured to show data to a user while the I/O device **320** may be a hardware component that enables the user to enter inputs. The display device **315** and the I/O device **320** may be separate components or integrated together such as a touchscreen. The transceiver **325** may be a hardware component configured to establish a connection with the 5G NR-RAN **120**, an LTE-RAN (not pictured), a legacy RAN (not pictured), a WLAN (not pictured), etc. Accordingly, the transceiver **325** may operate on a variety of different frequencies or channels (e.g., set of consecutive frequencies).

(28) As mentioned above, in a first aspect, the exemplary embodiments relate to beam reporting for multi-TRP operation. FIG. **4** shows a signaling diagram **400** for an exemplary multi-TRP beam reporting procedure according to various exemplary embodiments. The signaling diagram **400** is described with regard to the network arrangement **100** of FIG. **1** and the UE **110** of FIG. **3**.

(29) For beam reporting, a TRP may transmit multiple beams to the UE **110**. Each beam may include reference signals that are to be measured by the UE **110**. The UE **110** may then select a beam and report the selection to the network. In response, the network may send downlink data to the UE **110** via the selected beam. The example provided below will describe a beam reporting procedure for multi-TRP operation that includes implementing various exemplary beam management techniques on the UE **110** side and on the network side.

(30) The signaling diagram **400** includes the UE **110** and the cell **120A**. As mentioned above, the cell **120A** may represent a gNB configured with multiple TRPs. In some embodiments, the TRPs may be deployed locally at the gNB. In other embodiments, the TRPs may be deployed at various different locations and connect to the gNB via a backhaul connection. An example of this distributed arrangement is shown in FIG. **2**. The signaling diagram **400** is described with regard to a first TRP and a second TRP. However, the exemplary embodiments are not limited to two TRPs, those skilled in the art will understand how the exemplary concepts described herein may apply to any appropriate number of TRPs.

(31) In **405**, the cell **120A** may configure multiple groups of reference signals and each group may correspond to a single TRP. For example, a first group of reference signals may be configured for transmission by a first TRP and a second group of reference signals may be configured for transmission by a second TRP. Throughout this description, the term “group pair” may refer to a first group of reference signals configured for transmission by a first TRP that are associated with a second group of reference signals configured for transmission by a second TRP.

(32) For group reporting, the cell **120A** may constrain the group pair to reflect whether the cell

120A can transmit at the same time. For example, a gNB may not be able to transmit multiple beams from the same TRP simultaneously. However, the gNB may be able to transmit different beams from different TRPs simultaneously.

(33) A group of reference signals may refer to a group of channel state information reference signals (CSI-RS), a group of synchronization signal blocks (SSBs) or a group of any other appropriate type of reference signals. FIG. 5 illustrates an example of a CSI-RS group pair. In this example, a first TRP **505** performs a sweep using a first group of CSI-RS **510** that includes CSI-RS **511-514** and a second TRP **520** performs a sweep using a second group of CSI-RS **525** that includes CSI-RS **526-529**. The first group of CSI-RS **510** and the second group of CSI-RS **525** are associated with one another to form a group pair. As will be described in more detail below, in this type of scenario, the UE **110** may collect measurement data corresponding to each group **510**, **525**. The UE **110** may then select one of CSI-RS **511-514** from the first group **510** and one of CSI-RS **526-529** from the second group **525**. Subsequently, the first TRP **505** may be configured to provide the UE **110** with downlink data using a beam corresponding to the CSI-RS selected from the first group **510** and the second TRP **520** may be configured to provide the UE **110** with downlink data using a beam corresponding to the CSI-RS selected from the second group **525**.

(34) Returning to the signaling diagram **400** of FIG. 4, in some embodiments, a CSI-RS group may be based on a non-zero power (NZP)-CSI-RS-ResourceSet. As indicated above, each NZP-CSI-RS-ResourceSet may correspond to one TRP. In some embodiments, each NZP-CSI-RS-ResourceSet may correspond to a different TRP. In other embodiments, each NZP-CSI-RS-ResourceSet may correspond to the same TRP. In addition, for an NZP-CSI-RS-ResourceSet, multiple NZP-CSI-RS-Resources can be configured belonging to different groups.

(35) For a group of either NZP-CSI-RS-ResourceSet or NZP-CSI-RS-Resource, the following configurations may be utilized. In some embodiments, each NZP-CSI-RS-ResourceSet or NZP-CSI-RS-Resource may be configured with a physical cell ID (PCI). In some embodiments, each NZP-CSI-RS-ResourceSet or NZP-CSI-RS-Resource may be configured with a CORESETPoolIndex. In some embodiments, each NZP-CSI-RS-ResourceSet or NZP-CSI-RS-Resource may be configured with a separate group index. If the group configuration is missing, the UE **110** may assume that the group is from the first TRP (e.g., index 0).

(36) Prior to the transmission of the group pair, the network may provide the UE **110** with a CSI-ReportConfig message that is configured to indicate to the UE **110** how to report various types of CSI. When group based reporting is configured, the network may include group based reporting information in the CSI-ReportConfig message or any other appropriate message.

(37) In some embodiments, the CSI-ReportConfig message may indicate a bitmap based on the CSI-RS resource level. To provide an example, for (N) CSI-RS or SSB, a (N) bitmap may be configured. For each bit, a 0 may indicate that the corresponding CSI-RS or SSB belongs to the first TRP and a 1 may indicate that the corresponding CSI-RS or SSB belongs to the second TRP. In some embodiments, the CSI-ReportConfig message may indicate segmentation at the resource level. To provide an example, for (N) CSI-RS or SSB, the gNB may configure a single value (M) $< (N)$, the first (M) CSI-RS resources may correspond to the first TRP and the remaining N-M CSI-RS resources may correspond to the second TRP. The examples provided above were described at the CSI-RS resource level, however, those skilled in the art will understand that the above examples may also be applies to the CSI-RS resource set level.

(38) When multiple groups of resources are configured, the gNB may use the CSI-ReportConfig message or any other appropriate message to configure the UE **110** to select a beam based on particular criteria. One exemplary criterion instructs the UE **110** to perform beam selection on one group independently of the other group. In other words, the selection of a beam from one group is not correlated to the selection of a beam from the other group. This allows the UE **110** maximize the received reference signal received power (RSRP) or signal-to-interference-to-noise ratio (SINR). Another exemplary criterion instructs the UE **110** to jointly select beams in different

resource groups. In other words, the UE **110** may consider the mutual interference of both beams.

(39) In **410**, the cell **120A** may transmit a first group to the UE **110** via a first TRP. In **415**, the cell **120A** may transmit a second group to the UE **110** via a second TRP.

(40) In **420**, the UE **110** may collect measurement data corresponding to each group. For example, the UE **110** may collect measurement data corresponding to each CSI-RS or SSB of the first group and measurement data corresponding to each CSI-RS or SSB of the second group.

(41) In **425**, the UE **110** may select one beam from each group. The selection may be performed in accordance with information received from the network, the collected measurement data and/or any other appropriate factors.

(42) In **430**, the UE **110** may report the beam selection to the network. In some embodiments this may include transmitting information to the gNB. In other embodiments, this may include performing an uplink transmission to the TRPs using an uplink resource associated with the selected beam. However, these examples are merely provided for illustrative purposes, the UE **110** may report the beam selection and/or the corresponding measurement data to the network in any appropriate manner.

(43) FIG. **6** shows a signaling diagram **600** for an exemplary beam failure procedure for multi-TRP operation according to various exemplary embodiments. The signaling diagram **600** will be described with regard to the network arrangement **100** of FIG. **1** and the UE **110** of FIG. **2**.

(44) The signaling diagram **400** described various examples of how the UE **110** may acquire multiple beams from different TRPs. The signaling diagram **600** will describe various examples of how beam failure recovery may be performed when the UE **110** is configured with multiple beams from multiple TRPs. The signaling diagram **600** includes the UE **110** and the cell **120A**.

(45) As mentioned above, the cell **120A** may represent a gNB configured with multiple TRPs. In some embodiments, the TRPs may be deployed locally at the gNB. In other embodiments, the TRPs may be deployed at various different locations and connect to the gNB via a backhaul connection. An example of this distributed arrangement is shown in FIG. **2**. The signaling diagram **600** is described with regard to a first TRP and a second TRP. However, the exemplary embodiments are not limited to two TRPs, those skilled in the art will understand how the exemplary concepts described herein may apply to any appropriate number of TRPs.

(46) In **605**, the UE **110** acquires a first set of beams from a first TRP. In **610**, the UE **110** acquires a second set of beams from a second TRP. In **615**, the UE **110** monitors each beam for an indication of beam failure.

(47) For multi-TRP operations, beam failure may be triggered for each TRP. Thus, each set of beams may include reference signals (e.g., Radio Link Monitoring RS) configured to beam failure detection. The UE **110** may collect measurement data corresponding to these reference signals and declare beam failure when a predetermined condition is met (e.g., one or more measurement values fall below a predetermined threshold, etc.).

(48) On the network side, for multi-downlink control information (multi-DCI) based multi-TRP, to configure the per TRP Radio Link Monitoring RS, CORESETPoolIndex or PCI can be configured wot each Radio Link Monitoring RS or a group of Radio Link Monitoring RS. In some embodiments, when Radio Link Monitoring RS is not configured for a TRP, the UE **110** may monitor the beam used for physical downlink control channel (PDCCH) reception for the CORESET configured with the corresponding CORESETPoolIndex or PCI.

(49) In **620**, the UE **110** identifies one or more beam failure events. In **625**, the UE **110** transmits a beam failure recovery request to the cell **120A** indicating that one or more beam failure events have occurred.

(50) If the beam failure event at a TRP that is operating as a primary cell (PCell), different physical random access channel (PRACH) sequences/configuration may be used to indicate which TRP has beam failure. To provide an example, for a contention free RACH (CFRA) based beam failure request, a radio resource control (RRC) or medium access channel control element (MAC-CE) may

be used to indicate which TRP has beam failure. If the beam failure event at a TRP that is operating as a secondary cell (SCell), a MAC-CE may be used to indicate which TRP has beam failure.

(51) Separate candidate beams may be configured for each TRP. For intra-cell multi-TRP, CORESETPoolIndex may be used to indicate and configure the candidate beam for each TRP. For inter-cell multi-TRP, PCI can be used to indicate and configure the candidate beam for each TRP. This candidate information may be provided by the UE **110** in the beam failure recovery request or in any other appropriate message.

(52) In **630**, the cell **120A** may transmit a beam failure response to the UE **110**. In some embodiments, a separate search space may be configured for each TRP. For intra-cell multi-TRP, CORESETPoolIndex may be used to indicate and configure the recovery search space for each TRP. For inter-cell multi-TRP, PCI may be used to indicate and configure the recovery search space for each TRP.

(53) In some embodiments, after the UE **110** triggers the beam failure request for a particular TRP, the UE **110** may automatically update the beam of the following channels based on the UE **110** indicated candidate beam. For downlink reception, the channels may include the physical downlink shared channel (PDSCH) and the PDCCH. For uplink transmission, the channel include physical uplink control channel (PUCCH) and physical uplink shared channel (PUSCH).

(54) In some embodiments, the UE **110** may only update the beam in the corresponding TRP that the UE **110** triggers beam failure. For intra-cell multi-TRP, the UE **110** may indicate which TRP using the corresponding CORESETPoolIndex. For inter-cell multi-TRP, the UE **110** may indicate which TRP using the corresponding PCI.

(55) During operation, when the UE **110** is configured with multi-TRP, the UE **110** may be configured to receive overlapping CORESET in the time domain with different quasi co-located (QCL)-type D. In this type of scenario, if the UE **110** has previously reported the corresponding transmission indicator (TCI), the UE **110** may be monitoring two CORESETs simultaneously. In some embodiments, further restrictions may be imposed in the time domain (e.g. in the past (x) slots or milliseconds (ms)). Other indicators that may be monitored and reported may include one or more of reference signal indicators such as a CSI-RS Resource Indicator (CRI), a SS/PBCH Resource Block Indicator (SSBRI), a scheduling request indicator (SRI) or a sounding reference signal (SRS).

(56) Otherwise, the UE **110** monitors the CORESET with the beam from the highest priority CORESET. In this type of scenario, the UE **110** may or may not monitor the other CORESETs with lower priority. The priority may be determined based on the following factors, CORSET index, CORSETPoolIndex, SearchSpace index, Search Space periodicity, PCI, service cell index, or any other appropriate factor.

(57) In some embodiments, there may be a collision of different channels overlapping in the time domain with different QCL-typeD reception (e.g., PDCCH and PDSCH, PDSCH and PDSCH, PDSCH and CSI-RS, CSI-RS and CSI-RS. In this type of scenario, priority may be defined to determine which beam to use and the skipping of some channels. To provide an example, SSB may be the highest priority, CORESET may be the second highest priority, high priority downlink grant (DG)-PDSCH may be the third highest priority, high priority semi-persistent scheduling (SPS)-PDSCH may be the fourth highest priority, aperiodic CSI-RS may be the second fifth priority, low priority DG-PDSCH may be the sixth highest priority, low priority SPS-PDSCH may be the seventh highest priority, semi-persistent-PDSCH may be the eighth highest priority and periodic CSI-RS may be the lowest priority. However, the above example is merely provided for illustrative purposes, the priority order may be configured in any appropriate manner.

(58) Alternatively, for single-DCI based Multi-TRP, to configure the per TRP Radio Link Monitoring RS, PCI or any other appropriate logic index may be configured for each Radio Link Monitoring RS or a group of Radio Link Monitoring RS. When Radio Link Monitoring RS is not configured for a TRP, the UE **110** may monitor the beam used for PDCCH reception for the

CORESET configured with the corresponding PCI.

(59) Those skilled in the art will understand that the above-described exemplary embodiments may be implemented in any suitable software or hardware configuration or combination thereof. An exemplary hardware platform for implementing the exemplary embodiments may include, for example, an Intel x86 based platform with compatible operating system, a Windows OS, a Mac platform and MAC OS, a mobile device having an operating system such as iOS, Android, etc. The exemplary embodiments of the above described method may be embodied as a program containing lines of code stored on a non-transitory computer readable storage medium that, when compiled, may be executed on a processor or microprocessor.

(60) Although this application described various embodiments each having different features in various combinations, those skilled in the art will understand that any of the features of one embodiment may be combined with the features of the other embodiments in any manner not specifically disclaimed or which is not functionally or logically inconsistent with the operation of the device or the stated functions of the disclosed embodiments.

(61) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(62) It will be apparent to those skilled in the art that various modifications may be made in the present disclosure, without departing from the spirit or the scope of the disclosure. Thus, it is intended that the present disclosure cover modifications and variations of this disclosure provided they come within the scope of the appended claims and their equivalent.

Claims

1. An apparatus of a base station, the apparatus comprising processing circuitry configured to: configure transceiver circuitry to transmit a first group of reference signals based on a first non-zero power (NZP)-channel state information reference signal (CSI-RS) resource set using a first transmission reception point (TRP); configure transceiver circuitry to transmit a second group of reference signals based on a second different NZP-CSI-RS resource set using a second TRP, wherein the first group of reference signals are associated with the second group of reference signals as a group pair that is to be transmitted simultaneously; and decode, based on signals received from a user equipment (UE), an indication that the UE has selected one beam from each group of reference signals in the group pair.
2. The apparatus of claim 1, wherein each group of reference signals includes at least one CSI-RS.
3. The apparatus of claim 1, wherein the base station constrains group pairs based on whether the base station is configured to transmit multiple beams from a same TRP simultaneously.
4. The apparatus of claim 1, wherein the processing circuitry is further configured to: configure transceiver circuitry to transmit a channel state information (CSI) report configuration message to the UE, the CSI report configuration message comprising configuration information for group based reporting.
5. The apparatus of claim 4, wherein the CSI report configuration message includes a value configured to indicate that the first group of reference signals is transmitted by the first TRP and the second group of reference signals is transmitted by the second TRP.
6. A baseband processor configured to perform operations comprising: generating a first group of reference signals having only one or more channel state information reference signal (CSI-RS) and one or more synchronization signal block (SSB) to be transmitted using a first transmission reception point (TRP); generating a second group of reference signals having only one or more

CSI-RS and one or more SSB to be transmitted using a second TRP, wherein the first group of reference signals are associated with the second group of reference signals as a group pair that is to be transmitted simultaneously; and receiving an indication that a user equipment (UE) has selected one beam from each group of reference signals in the group pair.

7. The baseband processor of claim 6, wherein a base station constrains group pairs based on whether the base station is configured to transmit multiple beams from a same TRP simultaneously.

8. The baseband processor of claim 6, wherein the one or more CSI-RS of the first group of reference signals is a non-zero power (NZP)-CSI-RS and the one or more CSI-RS of the second group of reference signals is a NZP-CSI-RS.

9. The baseband processor of claim 6, the operations further comprising: transmitting a channel state information (CSI) report configuration message to the UE, the CSI report configuration message comprising configuration information for group based reporting.

10. The baseband processor of claim 9, wherein the CSI report configuration message includes a value configured to indicate that the first group of reference signals is transmitted by the first TRP and the second group of reference signals is transmitted by the second TRP.

11. A method, comprising: at a cell of a network: transmitting a first group of reference signals based on a non-zero-power (NZP) channel state information reference signal (CSI-RS) using a first transmission reception point (TRP) to a user equipment (UE); transmitting a second group of reference signals having only one or more channel state information reference signal (CSI-RS) and one or more synchronization signal block (SSB) using a second TRP to the UE, wherein the first group of reference signals are associated with the second group of reference signals as a group pair that is to be transmitted simultaneously; and receiving an indication that the UE has selected one beam from each group of reference signals in the group pair.

12. The method of claim 11, wherein the cell constrains group pairs based on whether the cell is configured to transmit multiple beams from a same TRP simultaneously.

13. The method of claim 11, the operations further comprising: transmitting a channel state information (CSI) report configuration message to the UE, the CSI report configuration message comprising configuration information for group based reporting.
